Modeling Cougar Predation Patterns: Leveraging Accelerometer Data for Enhanced Behavioral Insights

Matthew Pacey
Electrical Engineering and Computer Science
Oregon State University
paceym@oregonstate.edu

Abstract

We present a novel method to improve cougar predation modeling in the Northern Range of Yellowstone National Park to aid wildlife researchers in tracking the interactions between apex predators and prey. A sample of the cougar population was fit with global positioning system (GPS) telemetry collars to estimate the number of deer and elk they kill. The current estimation methods rely solely on GPS data in conjunction with determining the prey cause of death based on tracks and other visual signs. While effective, these methods cannot always differentiate between predation events—where the cougar actively kills prey—and scavenging events, such as when the collared cougar feeds on a carcass killed by another cougar. Our novel approach uses tri-axial accelerometer data from the collars to predict fine-scale behaviors. These behaviors can help distinguish predation events from scavenging events, providing wildlife researchers with more accurate predation rates and thus deeper insights into cougar-prey relationships.

1 Introduction

Predation rate is a critical metric for wildlife researchers to understand the top-down effects predators exert on prey populations. This rate quantifies the proportion of a prey population killed by a predator population. When predation rates exhibit a density-dependent relationship with prey density, it suggests that predation could be regulating the prey population and thus, demonstrates the predator's role in the ecosystem [1]. A key aspect of predation rate is the kill rate, which measures the number of prey killed by an individual over a specific period of time.

While collecting these data pose significant challenges, the estimation of predation rates can be greatly improved by fitting predators with Global Positioning System (GPS) telemetry collars [2].By leveraging these GPS data in real time, researchers can visit aggregations of GPS fixes, also known as GPS clusters, shortly after the predator was there to search for prey remains. Researchers can then search for evidence on how the prey died, such as signs of struggle or disturbance, to assign cause of death. However, assigning cause of death can be challenging and subjective. Furthermore, even when cause of death can reasonably be assigned to a cougar predation event, researchers cannot be certain that it was in fact the focal cougar that made the kill. Thus, kill rates, and the subsequent predation rates, are subject to being overestimated.

For example, let's assume a hypothetical but plausible scenario with an elk population of 8,000 and cougar population of 40. A typical cougar will feed approximately four times a month which translates to 48 times a year. We want to calculate the predation rates in two different scenarios:

- **Scenario 1**: All cougar feeding events are treated as predation. The cougar population would kill 1,920 elk (48 feedings $\times$ 40 cougars), equating to a predation rate of 24% ($\frac{1,920}{8,000}$).

- **Scenario 2**: Assume our method exposes some feeding events as scavenge events, with one feeding a month being scavenged. This would reduce predation events to 1,440 (36 feedings $\times$ 40 cougars), resulting in a predation rate of 18% ($\frac{1,440}{8,000}$).

Such a difference in predation rate estimates could affect inferences made about a predator population's ability to regulate prey populations, warranting investigation. Tri-axial accelerometers can aid this investigation by providing a finer resolution view of an animal's movements, capturing data multiple times per second. In contrast to the coarse data from GPS clusters, accelerometer data can offer detailed, three-dimensional insights into the animal's fine-scale behaviors.

Therefore, we propose a novel method to estimate cougar predation by augmenting the current GPS methods with accelerometer data. GPS data will be used to identify feeding clusters, including the location and estimated start time of the feeding. Accelerometer data will then help researchers differentiate whether a particular feeding cluster was a predation or scavenging event by analyzing the fine-scale movement data of the cougar during the feeding window. Using previously labeled known kills, we train a model to recognize a specific sequence of behaviors characteristic of predation events (stalk, kill, feed). In the absence of these specific behaviors, we will classify the event as scavenging. This method aims to provide wildlife researchers with deeper insights into the feeding habits of cougars and their impact on the prey populations in Yellowstone.

2 Background

2.1 Cougar Behavior and Its Impact on the Yellowstone Ecosystem

Our research aims to deepen our understanding of cougar behavior and its impact on the Yellowstone ecosystem. Of particular interest are the effects of cougars on the population dynamics of ungulates, including elk, mule deer, and white-tailed deer.

Cougars are known to have expansive home ranges, typically spanning between 20 and 150 square miles. Within these large territories, localized movements and fidelity to a small geographic space often indicate feeding activity. This is because cougars often kill large ungulates that require multiple days of feeding to consume. Leveraging this knowledge, our GPS clustering algorithm identifies potential feeding clusters. Additionally, accelerometer data collected around the onset of these clusters (plus or minus six hours) provide valuable behavioral insights.

Our behavioral model capitalizes on the fact that cougars are spot-and-stalk predators, utilizing patterns in accelerometer data to discern hunting behaviors, as detailed in Section 2.3. While cougars are solitary hunters, instances of carcass sharing have been observed, and this is thought to occur more commonly as prey size increases [3]. A secondary benefit of our novel method is that it can provide more insight into how common this carcass sharing is in the Northern Range of Yellowstone.

All of these known attributes of cougars will be utilized in service of providing more accurate kill and predation rates. This knowledge will be leveraged in both the data collection and analysis.

2.2 Current Method: GPS

The current method for estimating predation involves using GPS data generated from the collared animal. The animal's position is reported via a GPS fix every three hours. If multiple fixes occur within a 200-meter area over a rolling six-day window, this is identified as a potential feeding cluster. Researchers then hike to this cluster to search for prey remains. If remains are found, they attempt to determine the cause of death. If the cause of death appears to be from a cougar, it is counted as a predation event; otherwise, it is classified as a scavenging event. An example of a visualization of GPS clusters for a collared cougar can be seen in Figure 1.

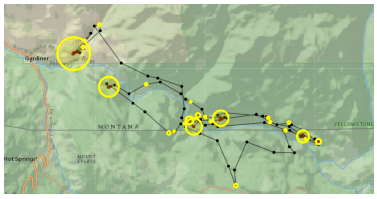


Figure 1: Example GPS cluster of a collared cougar. Each yellow circle represents a GPS cluster, indicating that the cougar spent at least three hours in that particular location.

2.3 Novel Method: Accelerometer

The GPS clusters are useful indicators of possible feeding locations, but they do not provide insights into how the collared animal came to feed on the carcass. Researchers aim to determine whether the event was a predation event (the cougar killed the prey) or a scavenging event (the collared cougar did not kill the prey). While GPS data cannot answer this question, the fine-grain detail offered by accelerometers can provide valuable insights.

The collars, manufactured by Vectronic, provide tri-axial acceleration data, meaning they report dynamic body acceleration in three dimensions (X, Y, and Z axes). These collars report data at 16 Hz, which translates to 48 readings of body acceleration per second (16 measurements per second along each of the three axes). Accelerometer data is continuously recorded from the time the animal is initially collared until the collar is retrieved, which can be up to two years later. Unlike GPS data, accelerometer data is not reported live; it can only be reviewed once the collar has been retrieved, either by recapturing the cougar or after the cougar dies.

2.4 Motion Activated Trail Cameras

Yellowstone National Park spans approximately 2.2 million acres [4], making it impossible to maintain constant visibility on these predators. However, motion-activated trail cameras provide valuable data. These cameras are deployed both in permanent locations and istrategically placed locations as needed. The data collected by these cameras can help label specific behaviors, as detailed in Section 4.1.

3 Technical Approach

Our approach to differentiating between predation and scavenging events comes in three phases: preparation, processing, and analyzing. The preparation of the data is done by assigning behaviors to specific sections of the data by leveraging data previous research and new data collection. During processing, specific sections of the accelerometer data is labeled with a specific behavior code. The processing is done using an open-source toolkit called the Bio-logger Ethogram Benchmark (BEBE)[5] which makes behavioral predictions of input data. Finally, we analyze the behavioral predictions using domain knowledge to look for indications of predation in the behavioral predictions. We will cover each of these three phases in greater detail here.

4 Technical Approach

Our approach to differentiating between predation and scavenging events consists of three phases:

1. **Data Preparation:** Behaviors are assigned to specific sections of the data by leveraging both previous research and new data collection.
2. **Data Modeling:** A model is trained on the labeled data, to make predictions on new data.
3. **Inference:** Analyze the behavioral predictions using domain knowledge to identify indications of predation.

Each of these three phases will be discussed in greater detail in the following sections.

4.1 Data Preparation

Our dataset comprises observations from previous studies of Yellowstone cougars, supplemented with newly collected data from trail cameras. The primary objective of our data preparation phase is to assign behavior codes to specific segments of accelerometer data.

Previous Work
Through previous work of Colby Anton [6], we have 150 identified kills by collared cougars. This dataset consists of an estimated time of kill (to a one-minute resolution), type and size of prey, and accompanying accelerometer data. We are enhancing this dataset by labeling periods of interest to the fine-scale resolution of one second. For our behavioral model we are interested in labeling the following behaviors in the accelerometer timestamps:

- **Low Energy/Stalk:** This period occurs immediately before the kill, characterized by little to no movement in any of the three axes. Given that cougars are spot-and-stalk predators, this period corresponds to the animal either lying still or stalking with minimal movement.
- **High Energy/Kill:** This period corresponds to the cougar attacking the prey, characterized by high-energy spikes (readings consistently over 0.1 to 0.2 gs). With larger prey (elk/deer), the cougar may have attached itself to the animal's neck, resulting in significant movement visible as the animals struggle.
- **Feeding:** Initially, we speculated on feeding behavior within a window of up to 30 minutes after the kill, looking for consistent medium energy in the accelerometers (corresponding to the cougar pulling off pieces of the prey and chewing). However, late in the project, we obtained trail cam footage of cougars feeding and used that accelerometer data instead of speculative feeding estimates. Our dynamic approach allows researchers to train and evaluate models on either approach.

An example visualization of labeled accelerometer data is shown in Figure 2. The data is labeled in a spreadsheet available at this link. Each collared individual has a detailed entry for each kill, including the start and end times for periods of interest: stalking, killing, and feeding. The data also includes info that is currently unused but may be used in later, more sophisticated models, such as the maternal status of the cougar and prey species and age.

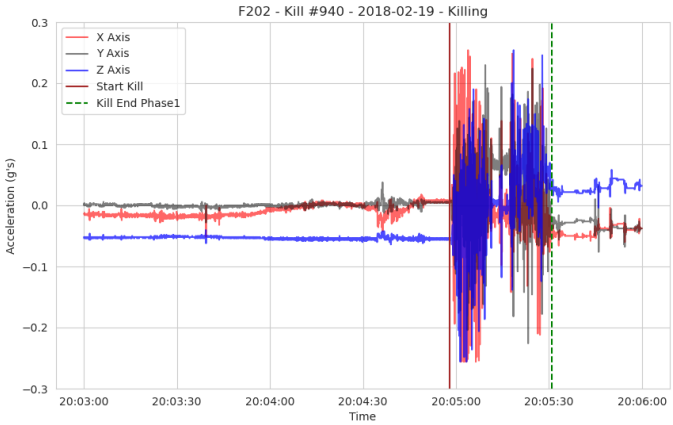


Figure 2: A plot of accelerometer data for a known cougar kill, with all three axes of the accelerometer displayed together in a semi-transparent format to ensure that any overlap still reveals all the data. The stalking behavior occurs from the start of the plot to the start of the kill, marked with a vertical dark red line at 20:04:28. The end of the kill, where absolute values no longer exceed 0.1 gs, is marked with a vertical dotted green line at 20:05:31.

Trail Cams

We are using two variations of trail cameras to label behaviors of interest:

- **Static** trail cameras help us label walking behavior because we can identify collared individuals in the videos and assume that they were walking at the time (thus triggering the camera with their movement). By checking the motion and video logs, we are able to pinpoint the exact times the cougar is walking and label the corresponding accelerometer data.
- **Dynamic** cameras allow for capturing feeding in situ. Yellowstone researchers identified feeding clusters and hiked out during the day to find carcasses that were only partially consumed. They deployed trail cameras pointed at the carcasses and collected numerous videos of animals feeding on the carcasses, including the collard cougars. Using timestamps from the trail cam footage, we translated these to timestamps in the accelerometer data and labeled the data accordingly.

4.2 Behavioral Modeling from Labeled Data

We use the Bio-logger Ethogram Benchmark (BEBE) [5] to make behavioral predictions from the labeled data. BEBE is an open-source tool specifically designed for processing accelerometer data from a wide range of animals. By providing a standardized task for behavior classification and corresponding evaluation metrics, BEBE facilitates comparisons between different machine learning techniques, aiding in the identification of best practices and promising directions for future research. Using BEBE, we process the accelerometer data to classify behaviors such as walking, stalking, killing, and feeding, enabling us to distinguish between predation and scavenge events more accurately. A key feature of the tool is how easy it makes it to evaluate competing machine learning modeling techniques with no extra overhead, saving us development time. Our process in using BEBE involves the following steps:

1. **Preprocess labeled data:** Our labeled data starts at 16Hz but our system allows downsampling of this data. The data is downsampled by averaging X input values into one output value. For example, if X is 2 our output is 8Hz.
2. **Select BEBE hyperparameters:** BEBE is configured to process data from many different types of wildlife, so some data-specific adjustments were needed. The most significant was increasing the size of the temporal window samples, also known as the clip size. To speed up processing, each input accelerometer segment is chopped up into clips of 2048 samples each. Behavioral predictions are made for each time point in these clips, using only data in these clips. The clips are then reassembled at the end. We found that by doubling the clip size, we achieved more accurate predictions. We also made minor tweaks to reduce runtime, such as lowering the number of epochs from 100 to 50 as testing accuracy converged by then.
3. **Select model type:** The BEBE tool allows for easy evaluation of different model types to find the best method for the given data type. For our experiments, we used a one-dimensional convolutional neural network (CNN), a convolutional-recurrent neural network (CRNN), random forests (RF), and decision trees (DT). We found the best results in CRNNs, as detailed in Section 5.2.
4. **Train and run model:** BEBE runs five-fold validation on the input, using a grid of hyperparameters. For each run, it uses four of the folds for training and the held-out fold for testing. The model that has the highest F1 score (averaged across each fold) for behavioral prediction accuracy is chosen. We take the behavioral predictions from this model to use in our analysis.

4.3 Inference

We utilize domain knowledge from a subject matter expert [7] to define rules for filtering behavioral predictions. According to these rules, a feeding event is classified as predation if it follows a specific sequence: low energy/stalk, high energy/kill, and feeding, all within a predefined time window. If this sequence is identified, the feeding period is classified as a predation event; otherwise, it is classified as a scavenging event.

Our check predicted kills script encodes our domain knowledge into an easily configurable package. For a given chunk of accelerometer data, it must meet all the criteria below to be considered a predation event. If any of these criteria are not met, the event will be considered a scavenging event. For each item, we include our selected value. Future iterations can explore and expand these values as needed.

- **Stalk offset:** Stalking behavior must occur within a specific window immediately prior to the start of kill behavior. We chose 5 seconds, as in there must be at least one instance of stalking behavior before the first kill behavior.
- **Stalk duration:** The stalking behavior must last at least a minimum duration. We selected a duration of 5 seconds.
- **Kill time:** The duration of kill behavior must exceed a minimum length to avoid mistaking short high-activity bursts for kill indicators. We selected 3 seconds.

- **Feed Start:** Feeding behavior must begin within a specific window after the end of kill behavior. We chose 1 hour, as in there must be at least one instance of feeding behavior after the last kill behavior. Future iterations may benefit from adjusting this for mothers with kittens to account for the fact that mothers with kittens may retrieve their kittens before feeding, which could take several hours.
- **Feed Window:** This is the time window to accumulate feeding behavior. After the kill has ended, we will count feeding during this period. This ensures that each feeding event has a consistent time window to search for feeding behaviors. We selected 3 hours, meaning that any feeding found up to three hours after the kill is accumulated for the feed duration.
- **Feed Duration:** The cumulative feeding time within the feeding window must exceed a minimum duration. This duration can include breaks for other behaviors, such as walking, between feeding bouts. We started with 5 minutes, this may be increased later as behavioral prediction accuracy increases.

An example visualization can be seen in Figure 3. This plot shows three distinct kill behaviors, but only one met the criteria to be considered a kill event according to our filtering rules.

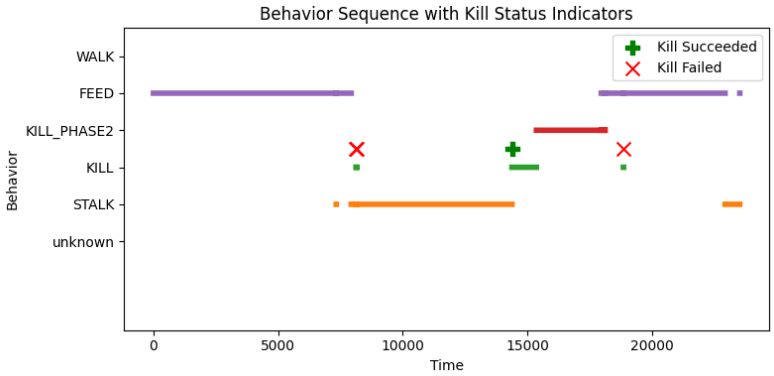


Figure 3: Example visualization of domain knowledge filtering plot. Each behavior of interest is represented on a separate row and color-coded. For each kill behavior (green), our domain knowledge filtering determines if it corresponds to a kill event (green plus) or not (red X).

5 Results

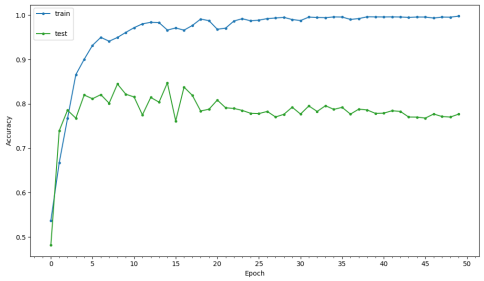
5.1 Behavioral Prediction

The accuracy of our behavioral model is evaluated by comparing predicted behaviors against labeled behaviors. When labeling our kill data, we only labeled a portion of the input data, as we were interpreting accelerometer data without accompanying video. Consequently, a significant portion of the input data remains unlabeled, as illustrated in Figure 5a (first row). The 'Ground Truth Behavior Labels' section shows our labels, with many timesteps labeled as unknown in BEBE due to the absence of behavior data.

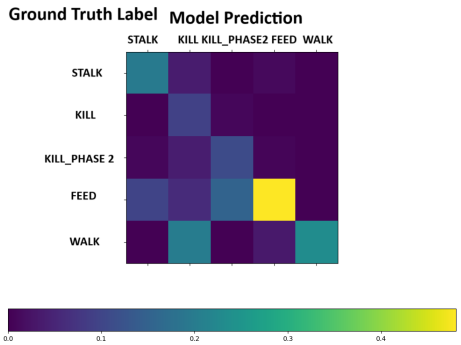
Labeling data is a labor-intensive process. To test the viability of our workflow, we initially labeled 5 kill sequences from 5 different individuals out of our 150 known kill events. This approach allowed us to validate the workflow on a small dataset before committing to more extensive labeling. The results from running a single dataset through BEBE can be seen in Figure 4a. The model converges to nearly perfect training accuracy and achieves 80% accuracy on the held-out test data.

A confusion matrix of the behavior predictions is shown in Figure 4b. This matrix displays the ground truth labels along the rows and the model predictions along the columns. The diagonal from the top left to the bottom right represents accurate predictions. Lighter colors indicate higher accuracies, while darker colors indicate lower accuracies.

In this confusion matrix, the 'feeding' label is the most accurately predicted behavior, with an accuracy of approximately 50%. In contrast, the 'kill' behavior is predicted with about 15% accuracy.



(a) Training and testing accuracy when running our data through a Convolutional-Recurrent Neural Network (CRNN). With a small dataset, the network achieved perfect training accuracy and a maximum of 80% accuracy on validation sets.



(b) Confusion Matrix of model predictions. The rows represent ground truth labels and columns represent model predictions. Lighter colored cells represent more accurate behavior predictions.

Figure 4: Comparison of Accelerometer Data and Confusion Matrix

Notably, when the ground truth label is 'walk', it is misclassified as 'kill' at a similar rate to 'kill' itself, highlighting an area for improvement in the model. To address this, increasing the number of labeled walking videos across more individuals should help enhance the model's accuracy in distinguishing between these behaviors.

5.2 Model Comparison

The BEBE tool facilitates the comparison of different modeling paradigms to identify the best fit for a given dataset. It supports both supervised and unsupervised models. For our purposes, we opted for supervised models since we have labeled our data and are interested in identifying specific behaviors rather than undirected groupings of behaviors.

We compared a Convolutional-Recurrent Neural Network (CRNN) against a Random Forest (RF) approach. The results of this comparison are shown in Figure 5. Both models predicted the labeled behaviors reasonably well; however, the RF model's predictions were much choppier, with frequent switches between behaviors. In contrast, the CRNN model provided smoother and more consistent predictions. Based on these observations, we plan to continue using the CRNN model in future iterations, especially as we gather more labeled data.

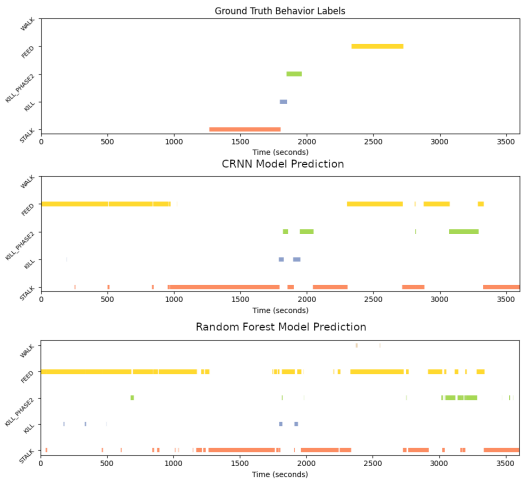
6 Summary

Our goal was to treat this project as a proof of concept to explore the plausibility of determining predation using accelerometer data. Given the encouraging results from the small-scale BEBE experiments, this was a success. With minimal examples labeled, the model predicted behaviors with enough accuracy to warrant future efforts into labeling. The addition of labeled feeding behavior from trail cams should also aid in this endeavor.

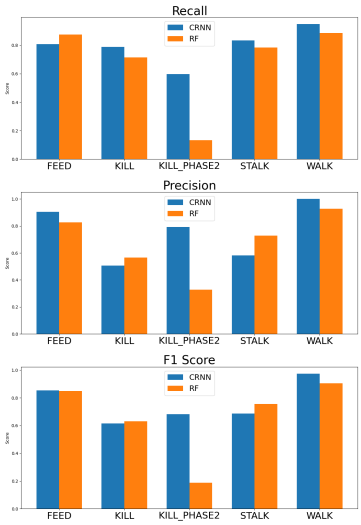
7 Advice for future students

7.1 Project Specific

- **Be aware of time zones and time drift:** The collar data is stored in Universal Time Coordinated (UTC) and trail cam timestamps are in Mountain Standard Time (MST). When labeling accelerometer data from trail cam footage, be sure to double check the conversion before including data in your model.
- **Train a model to identify prey type:** The source dataset contains prey species for each kill, so a refined model may be able to be trained to not only identify predation events but also to identify the type of prey.



(a) The ground truth labels are shown in the first row, CRNN behavioral predictions in the second row, and RF behavioral predictions in the third row.



(b) Cumulative performance metrics of each model across all five folds of training and testing. The recall, precision, and F1 scores are calculated for each behavior.

Figure 5: Comparison of CRNN and RF models using the same input data. Behavioral predictions of a single event on the left (a) and test scores over all data on the right (b).

7.2 General

- **Meet regularly with your project partner:** I had a weekly meeting with my project partner to track and plan short term goals. This was extremely helpful to ensure that the project was staying on track. It was also a great opportunity to answer any outstanding questions. An instantaneous back and forth helped build domain knowledge.
- **Value your project partner's time:** Between meetings, maintain a list of open questions. This will ensure that you come prepared to discuss blocks you have encountered. It may also be that you eventually answer the question yourself before the meeting and do not need to discuss in the meeting.
- **Utilize Office Hours:** You may have regular meetings with the Capstone lead, but those meetings are quite short. Take advantage of office hours and you may find time to dive deeper into questions and potential solutions during this time.

References

[1] Anthony R. E. Sinclair and Charles J. Krebs. "Mammal population regulation, keystone processes, and ecosystem dynamics". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 358.1438 (2003), pp. 1729–1740. DOI: 10.1098/rstb.2003.1359. URL: https://royalsocietypublishing.org/doi/10.1098/rstb.2003.1359.

[2] Håkan Sand et al. "Using GPS Technology and GIS Cluster Analyses to Estimate Kill Rates in Wolf-Ungulate Ecosystems". In: *Wildlife Society Bulletin* 33.3 (2005), pp. 914–925. DOI: 10.2193/0091-7648(2005)33[914:UGTAGC]2.0.CO;2.

[3] L. Mark Elbroch et al. "Adaptive social strategies in a solitary carnivore". In: *Science Advances* 3.10 (2017), e1701218. DOI: 10.1126/sciadv.1701218. URL: https://advances.sciencemag.org/content/3/10/e1701218.

[4] National Park Service. *Yellowstone National Park - Basic Information*. https://www.nps.gov/yell/planyourvisit/basicinfo.htm. Accessed: May 25, 2024.

[5] Benjamin Hoffman et al. *A benchmark for computational analysis of animal behavior, using animal-borne tags*. 2024. arXiv: 2305.10740 [cs.LG].

[6] C. Anton. "The Demography and Comparative Ethology of Top Predators in a Multi-Carnivore System". PhD thesis. UC Santa Cruz, 2020. URL: https://escholarship.org/uc/item/3xd9379b.

[7] Wesley Binder. "Apex Predator Interactions and Population Dynamics in Yellowstone National Park". PhD Thesis. Department of Fisheries, Wildlife and Conservation Sciences, 2027.